ME551 Vibrations in Linear Systems

Lecture 3

Linear System Theory 2

Chap 1. Linear System Theory

1.1 Problem Formulation of Linear System in Time Domain: $D[x(t)] = f(t)$
1.2 System Classification. Superposition Principle.
1.3 Excitation and Response Characterization
1.4 Response to Initial Conditions
1.5 Response to Harmonic Excitations: Frequency Response Function
1.6 Response to Arbitrary Excitations by Laplace Transform
1.7 Generalized Functions. Response to Generalized Functions
1.8 Response to Arbitrary Excitations by Convolution Integral
1.9 **State Equations.** Response by the **Transition Matrix**
1.10 **Discrete**-Time Systems (optional)

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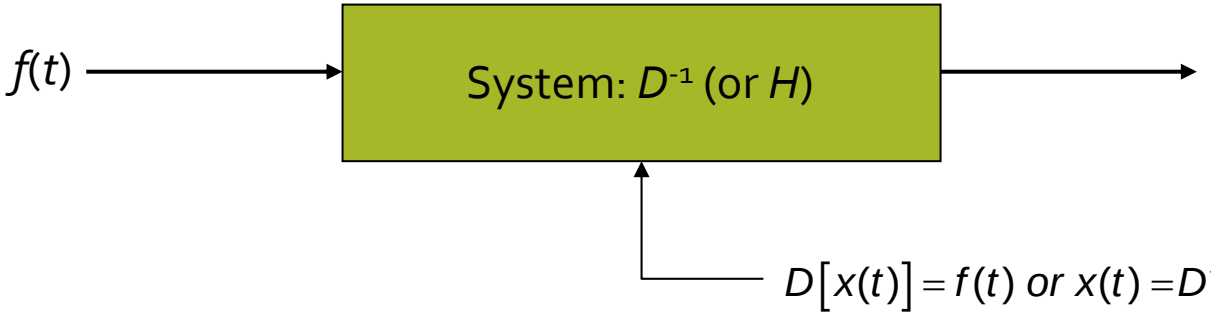
Recap:

1.1 Problem Formulation of Linear System in Time Domain: D[x(t)] = f(t) (or, x(t) = H[f(t)])

Problem Description:



$$D\left[x(t)\right] = f(t) \text{ or } x(t) = D^{-1}\left[f(t)\right] = H\left[f(t)\right]$$

D: Linear time-invariant differential operator

What if input f(t) and x(t) are n-by-1 vectors ?

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1.9 State Equations. Response by Transition Matrix (e^{At})
(1-st order SISO system versus higher order MIMO system)

SISO System: Single Input Single Output System
MIMO System: Multiple Input Multiple Output System

- **Impulse response by Laplace transform**

1-st order SISO (1-dim) system

$$\dot{x}(t) = ax(t) + f(t)$$

$$\downarrow\Leftarrow \text{L-transform under } x(0) = 0$$

$$X(s) = \underbrace{\frac{1}{s-a}}_{=G(s)} F(s)$$

$$x(t) = L^{-1}\left[G(s)F(s)\right]$$

$$\downarrow\Leftarrow \begin{cases} f(t) = \delta(t) \text{ with } x(-\varepsilon) = 0 \\ F(s) = 1 \end{cases}$$

$$g(t) = e^{at} : \text{ Unit impulse response}$$

(Actual examples?)

1-st order MIMO (n-dim) system

$$\underset{n\times 1}{\dot{\boldsymbol{x}}(t)} = \underset{n\times n}{\boldsymbol{A}}\,\underset{n\times 1}{\boldsymbol{x}(t)} + \underset{n\times r}{\boldsymbol{B}}\,\underset{r\times 1}{\boldsymbol{f}(t)}$$

$$\downarrow\Leftarrow \text{L-transform under } \boldsymbol{x}(0) = 0$$

$$[\boldsymbol{I}s - \boldsymbol{A}]\boldsymbol{X}(s) = \boldsymbol{B}\boldsymbol{F}(s):$$

$$\boldsymbol{X}(s) = \underbrace{[\boldsymbol{I}s - \boldsymbol{A}]^{-1}}_{\equiv \boldsymbol{G}(s):n\times n}\boldsymbol{B}\boldsymbol{F}(s)$$

$$\boldsymbol{x}(t) = L^{-1}\left[\boldsymbol{G}(s)\boldsymbol{B}\boldsymbol{F}(s)\right]$$

$$\downarrow\Leftarrow \textit{assume}, \boldsymbol{B}\boldsymbol{f}(t) = \delta(t) \text{ with } \boldsymbol{x}(-\varepsilon) = 0$$

$$\boldsymbol{g}(t) : \textit{impulse response vector}$$

Questions on terminology:

Order of system? Dimension of system ?
Dimension of state (vector) ?
Number of inputs? Number of outputs ?

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- ## Homogenous solutions with harmonic assumptions

1-st order SISO (1-dim) system

$$\dot{x}(t) = ax(t)$$

$$\downarrow \Leftarrow x(t) = Xe^{pt} \text{ : harmonic assumption}$$

$$Xpe^{pt} = aXe^{pt} : p = a$$

$$x(t) = Xe^{at}$$

$$\downarrow \Leftarrow \text{initial condition} : x(0)$$

$$X = x(0)$$

$$\downarrow$$

$$\therefore x(t) = x(0)e^{at}$$

$$\downarrow \Leftarrow when\ x(0) = 1$$

$$x(t) = g(t) = e^{at} : \text{Unit impulse response}$$

1-st order MIMO (n-dim) system

$$\underset{n\times n}{\dot{\boldsymbol{K}}}(t) = \underset{n\times n}{\boldsymbol{K}}(t)\,\underset{n\times n}{\boldsymbol{A}}$$

2) Transposition(See p.7)

$\downarrow$

$\Leftarrow$

$\uparrow$

1) $\underset{n\times n}{\boldsymbol{K}}(t) \equiv \left[\boldsymbol{x}_1(t) \quad \boldsymbol{x}_2(t) \quad \ldots \quad \boldsymbol{x}_n(t)\right]^T$:

Expansion to n–dim matrix state eq.

$$\begin{cases} \underset{n\times 1}{\dot{\boldsymbol{x}}}(t) = \underset{n\times n}{\boldsymbol{A}^T}\,\underset{n\times 1}{\boldsymbol{x}}(t) : \\ \text{one } n-\text{dim state eq.} \\ \quad \downarrow \Leftarrow \begin{cases} \text{generalization to } n-\text{equations} \\ n \text{ of } n-\text{dim state eqs.} \end{cases} \\ \underset{n\times 1}{\dot{\boldsymbol{x}}_i}(t) = \underset{n\times n}{\boldsymbol{A}^T}\,\underset{n\times 1}{\boldsymbol{x}_i}(t), \\ i = 1,2,\ldots,n \end{cases}$$

$$\downarrow \Leftarrow \begin{cases} \underset{n\times n}{\boldsymbol{K}}(t) = \underset{n\times n}{\boldsymbol{X}}\,\underset{n\times n}{e^{\boldsymbol{P}t}} \text{ : harmonic assumption} \\ \text{With Requirement of } e^{\boldsymbol{P}t} : \dfrac{d}{dt}\left(\underset{n\times n}{e^{\boldsymbol{P}t}}\right) = \underset{n\times n}{\boldsymbol{P}}\,\underset{n\times n}{e^{\boldsymbol{P}t}} = \underset{n\times n}{e^{\boldsymbol{P}t}}\,\underset{n\times n}{\boldsymbol{P}} \end{cases}$$

$$\underset{n\times n}{\boldsymbol{X}}\,\underset{n\times n}{\boldsymbol{P}}\,\underset{n\times n}{e^{\boldsymbol{P}t}} = \underset{n\times n}{\boldsymbol{X}}\,\underset{n\times n}{e^{\boldsymbol{P}t}}\,\underset{n\times n}{\boldsymbol{A}} \rightarrow \boldsymbol{P} = \boldsymbol{A}$$

$$\downarrow$$

$$\therefore \boldsymbol{K}(t) = \boldsymbol{X}e^{\boldsymbol{A}t} = \boldsymbol{K}(0)e^{\boldsymbol{A}t}$$

$$\downarrow \Leftarrow \text{When } \underset{n\times n}{\boldsymbol{K}}(0) = \underset{n\times n}{\boldsymbol{I}}$$

$$\boldsymbol{K}(t) = \boldsymbol{G}(t) = e^{\boldsymbol{A}t} : \text{ Transition matrix}\left(e^{\boldsymbol{A}t} \equiv ? \text{ for } \frac{d}{dt}\left(e^{\boldsymbol{A}t}\right) = \boldsymbol{A}e^{\boldsymbol{A}t} = e^{\boldsymbol{A}t}\boldsymbol{A}\right)$$

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- **Response to both initial condition and arbitrary forcing function by convolution integral**

1-st order SISO (1-dim) system

$$\dot{x}(t) = ax(t) + f(t)$$

Impulse response function e^{at} and initial condition $x(0)$

$$x(t) = \underbrace{e^{at}x(0)}_{\text{Response due to initial condition only}} + \underbrace{\int_0^t e^{a(t-\tau)}f(\tau)d\tau}_{\text{Response to forcing function only under zero initial condition}}$$

where $e^{at} \equiv 1 + \sum_{k=1}^{\infty} \frac{1}{k!} a^k t^k$

Cf) Taylor series: $y(x+h) = y(x) + \sum_{k=1}^{\infty} \frac{1}{k!} y^{(k)}(x) h^k$

1-st order MIMO (n-dim) system

$$\underset{n\times 1}{\dot{\boldsymbol{x}}(t)} = \underset{n\times n}{\boldsymbol{A}} \underset{n\times 1 \text{ vector}}{\boldsymbol{x}}(t) + \underset{n\times r}{\boldsymbol{B}} \underset{r\times 1}{\boldsymbol{f}}(t)$$

$e^{\boldsymbol{A}t}$ with initial vector $\boldsymbol{x}(0)$

$$\underset{n\times 1}{\boldsymbol{x}(t)} = \underset{n\times n}{e^{\boldsymbol{A}t}} \underset{n\times 1}{\boldsymbol{x}(0)} + \int_0^t \underbrace{e^{\boldsymbol{A}(t-\tau)}}_{n\times n} \underset{n\times r}{\boldsymbol{B}} \underset{r\times 1}{\boldsymbol{f}(\tau)}d\tau$$

where $\underset{n\times n}{e^{\boldsymbol{A}t}}$: Transition matrix

What is transition matrix $e^{\boldsymbol{A}t}$??

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- **Several expressions of 1-st order, MIMO (n-dim) system with vector ODE:**

$$\underset{n\times n}{\boldsymbol{B}}\ \underset{n\times 1}{\dot{\boldsymbol{x}}}(t) = \underset{n\times n}{\boldsymbol{C}}\ \underset{n\times 1}{\boldsymbol{x}}(t): \begin{cases} \text{one 1-st ode of } n\text{-dim vector variable } \boldsymbol{x}(t) \\ \text{with two } n\text{-dim coefficient matrices } \boldsymbol{B} \text{ and } \boldsymbol{C} \end{cases}$$

$$\Updownarrow$$

$$\underset{n\times 1}{\dot{\boldsymbol{x}}}(t) = \underbrace{\underset{n\times n}{\boldsymbol{B}^{-1}}\ \underset{n\times n}{\boldsymbol{C}}}_{\equiv \boldsymbol{A}^T}\ \underset{n\times 1}{\boldsymbol{x}}(t)$$

$$\Updownarrow$$

$$\underset{n\times 1}{\dot{\boldsymbol{x}}}(t) = \underset{n\times n}{\boldsymbol{A}^T}\ \underset{n\times 1}{\boldsymbol{x}}(t): \begin{cases} \text{one 1-st order order ode of } n\text{-dim vector variable } \boldsymbol{x}(t) \\ \text{with one } n\text{-dim coefficient matrix } \boldsymbol{A} \end{cases}$$

$$\Updownarrow$$

$$\begin{Bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ . \\ . \\ \dot{x}_n(t) \end{Bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & . & \dots & a_{2n} \\ a_{31} & . & . & \dots & . \\ . & . & . & \dots & . \\ a_{n1} & a_{n2} & . & \dots & a_{nn} \end{bmatrix}^T \begin{Bmatrix} x_1(t) \\ x_2(t) \\ . \\ . \\ x_n(t) \end{Bmatrix} : \begin{cases} n \text{ 1-st ode's of } n \text{ scalar variables } x_i(t) \\ \text{with } n^2 \text{ coefficients}, a_{ij}, i, j = 1, 2, \dots, n \end{cases}$$

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- **Expansion to 1-st order n-dim MIMO (*n*-dim) system with matrix ODE's:**

$$\underset{n\times 1}{\dot{\boldsymbol{x}}_i}(t) = \underset{n\times n}{\boldsymbol{A}^T}\,\underset{n\times 1}{\boldsymbol{x}_i}(t), i = 1,2,\ldots,n$$

$$\downarrow\Leftarrow \left\{ \underset{n\times n}{\boldsymbol{K}^T}(t) \equiv \begin{bmatrix} \boldsymbol{x}_1(t) & \boldsymbol{x}_2(t) & \ldots & \boldsymbol{x}_n(t) \end{bmatrix} = \begin{bmatrix} x_{11}(t) & x_{12}(t) & x_{13}(t) & \ldots & x_{1n}(t) \\ x_{21}(t) & x_{22}(t) & . & \ldots & x_{2n}(t) \\ x_{31}(t) & . & . & \ldots & . \\ . & . & . & \ldots & . \\ x_{n1}(t) & x_{n2}(t) & . & \ldots & x_{nn}(t) \end{bmatrix} \right.$$

$$\underset{n\times n}{\dot{\boldsymbol{K}}^T}(t) = \underset{n\times n}{\boldsymbol{A}^T}\,\underset{n\times n}{\boldsymbol{K}^T}(t):$$

$$\begin{bmatrix} \dot{x}_{11}(t) & \dot{x}_{12}(t) & \dot{x}_{13}(t) & \ldots & \dot{x}_{1n}(t) \\ \dot{x}_{21}(t) & \dot{x}_{22}(t) & . & \ldots & \dot{x}_{2n}(t) \\ \dot{x}_{31}(t) & . & . & \ldots & . \\ . & . & . & \ldots & . \\ \dot{x}_{n1}(t) & \dot{x}_{n2}(t) & . & \ldots & \dot{x}_{nn}(t) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} \\ a_{21} & a_{22} & . & \ldots & a_{2n} \\ a_{31} & . & . & \ldots & . \\ . & . & . & \ldots & . \\ a_{n1} & a_{n2} & . & \ldots & a_{nn} \end{bmatrix}^T \begin{bmatrix} x_{11}(t) & x_{12}(t) & x_{13}(t) & \ldots & x_{1n}(t) \\ x_{21}(t) & x_{22}(t) & . & \ldots & x_{2n}(t) \\ x_{31}(t) & . & . & \ldots & . \\ . & . & . & \ldots & . \\ x_{n1}(t) & x_{n2}(t) & . & \ldots & x_{nn}(t) \end{bmatrix}$$

$\downarrow\Leftarrow$ Transposition

$$\dot{\boldsymbol{K}}(t) = \boldsymbol{K}(t)\boldsymbol{A}$$

Have we increased information on the underlying system (A)?

Expansion from vector ode to matrix ode for what ?

- **How shall we obtain the transition matrix $e^{\boldsymbol{A}t}$? : Generalization of Taylor Series**

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & . & \dots & a_{2n} \\ a_{31} & . & . & \dots & . \\ . & . & . & \dots & . \\ a_{n1} & a_{n2} & . & \dots & a_{nn} \end{bmatrix}$$

$\downarrow$

Cf) Taylor series of exponential function:

$$e^{at} = 1 + \sum_{k=1}^{\infty} \frac{(at)^k}{k!}$$

$$(1)\ \underset{n\times n}{e^{\boldsymbol{A}t}} \equiv \underset{n\times n}{\boldsymbol{I}} + \sum_{k=1}^{\infty} \frac{1}{k!} \underset{n\times n}{\boldsymbol{A}^k} t^k \ (\text{Transition matrix}) \Rightarrow \frac{d}{dt}\left(\underset{n\times n}{e^{\boldsymbol{A}t}}\right) = \underset{n\times n}{\boldsymbol{A}}\ \underset{n\times n}{e^{\boldsymbol{A}t}} = \underset{n\times n}{e^{\boldsymbol{A}t}}\ \underset{n\times n}{\boldsymbol{A}}$$ Great !

or

$$(2)\ \underset{n\times n}{e^{\boldsymbol{A}t}} \equiv \begin{bmatrix} e^{a_{11}t} & e^{a_{12}t} & . & e^{a_{1n}t} \\ e^{a_{21}t} & e^{a_{22}t} & . & e^{a_{2n}t} \\ . & . & . & . \\ e^{a_{n1}t} & e^{a_{n2}t} & . & e^{a_{nn}t} \end{bmatrix} \Rightarrow \frac{d}{dt}\left(\underset{n\times n}{e^{\boldsymbol{A}t}}\right) = \begin{bmatrix} a_{11}e^{a_{11}t} & a_{12}e^{a_{12}t} & . & a_{1n}e^{a_{1n}t} \\ a_{21}e^{a_{21}t} & a_{22}e^{a_{22}t} & . & a_{2n}e^{a_{2n}t} \\ . & . & . & . \\ a_{n1}e^{a_{n1}t} & a_{n2}e^{a_{n2}t} & . & a_{nn}e^{a_{nn}t} \end{bmatrix} \neq \underset{n\times n}{\boldsymbol{A}}\ \underset{n\times n}{e^{\boldsymbol{A}t}}$$

→ Which one is the transition matrix do you think?

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Another derivation of 'Transition Matrix' in Linear Algebra (Eigen value Problem)

Transition matrix in this text: $\underset{n\times n}{e^{\boldsymbol{A}t}} \equiv \underset{n\times n}{\boldsymbol{I}} + \sum_{k=1}^{\infty} \underset{n\times n}{\boldsymbol{A}^k} \frac{t^k}{k!}$

Transition matrix in linear algebra: $\underset{n\times n}{\boldsymbol{A}}$ in $\left(\underset{n\times 1}{\dot{\boldsymbol{x}}} = \underset{n\times n}{\boldsymbol{A}}\,\underset{n\times 1}{\boldsymbol{x}}\right)$

The transition matrix $e^{\boldsymbol{A}t}$ can be calculated efficiently via Eigen value analysis on the matrix A.
See also Eigen value analysis in L12.

$$e^{\boldsymbol{A}t} = \boldsymbol{X} e^{\boldsymbol{\Lambda}t} \boldsymbol{Y}^T, \text{ where } e^{\boldsymbol{\Lambda}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \dots & \\ & & e^{\lambda_n t} \end{bmatrix} \Leftarrow \begin{cases} \text{Eigen analysis on } \boldsymbol{A} \text{ and } \boldsymbol{A}^T: \\ \boldsymbol{X} = [\boldsymbol{x}_1 \boldsymbol{x}_2 \dots \boldsymbol{x}_n] \text{ : right eigenvector} \\ \boldsymbol{Y} = [\boldsymbol{y}_1 \boldsymbol{y}_2 \dots \boldsymbol{y}_n] \text{ : left eigenvector} \\ \boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & & \\ & \dots & \\ & & \lambda_n \end{bmatrix} \end{cases} \Leftarrow \begin{cases} \boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x} \\ \boldsymbol{A}^T \boldsymbol{y} = \lambda \boldsymbol{y} \end{cases}$$

: Adjoint eigenvalue problems

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Calculation of Transition matrix and Matrix exponential in MATLAB:

$$1)\, e^{\boldsymbol{A}} \equiv \boldsymbol{I} + \sum_{k=1}^{\infty} \frac{\boldsymbol{A}^k}{k!}$$

$$\downarrow \Leftarrow \mathrm{expm}(\boldsymbol{A})$$

$$= \boldsymbol{V} * \mathrm{diag}(\exp(\mathrm{diag}(\boldsymbol{D})) * \boldsymbol{V}^T, \text{ where } [\boldsymbol{V}, \boldsymbol{D}] = eig(\boldsymbol{A})$$

: In case A is a symmetric matrix (i.e., X=Y=V)

$$2)\, e^{\boldsymbol{A}} \equiv \begin{bmatrix} e^{a_{11}} & e^{a_{12}} & \dots & e^{a_{1n}} \\ e^{a_{21}} & & & e^{a_{2n}} \\ . & & & \\ . & & & \\ e^{a_{n1}} & & & e^{a_{nn}} \end{bmatrix}$$

$$\downarrow \Leftarrow \exp(\boldsymbol{A})$$

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Any order of linear MIMO systems can be formulated into 1-st order *n*-dimensional ODE or represented by *n*-dim transition matrix.

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Example 1 : A SISO system of order l

$$\left(a_0\frac{d^l}{dt^l}+a_1\frac{d^{l-1}}{dt^{l-1}}+\ldots+a_l\right)x(t)=f(t)\Rightarrow\frac{d^l x}{dt^l}=-\frac{a_l}{a_0}x-\frac{a_{l-1}}{a_0}\frac{dx}{dt}-\ldots-\frac{a_1}{a_0}\frac{d^{l-1}x}{dt^{l-1}}+\frac{1}{a_0}f(t)$$

$$\downarrow\Leftarrow\begin{cases}x_1(t)\equiv x(t)\\ x_2(t)\equiv\dfrac{dx_1}{dt}=\dot{x}_1=\dfrac{dx}{dt}\\ x_3(t)\equiv\dfrac{dx_2}{dt}=\dot{x}_2=\dfrac{d^2x}{dt^2}\\ .\\ x_l(t)\equiv\dfrac{dx_{l-1}}{dt}=\dot{x}_{l-1}=\dfrac{d^{l-1}x}{dt^{l-1}}\end{cases}$$

$$\underbrace{\begin{Bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\\ .\\ \dot{x}_{l-1}(t)\\ \dot{x}_l(t)\end{Bmatrix}}_{\dot{\boldsymbol{x}}(t)}=\underbrace{\begin{bmatrix}0 & 1 & 0 & . & 0\\ 0 & 0 & 1 & . & 0\\ . & . & . & 1 & 0\\ 0 & 0 & . & 0 & 1\\ -\dfrac{a_l}{a_0} & -\dfrac{a_{l-1}}{a_0} & . & . & -\dfrac{a_1}{a_0}\end{bmatrix}}_{\boldsymbol{A}}\underbrace{\begin{Bmatrix}x_1(t)\\ x_2(t)\\ .\\ .\\ x_l(t)\end{Bmatrix}}_{\boldsymbol{x}(t)\text{: state vector}}+\underbrace{\begin{Bmatrix}0\\ 0\\ .\\ .\\ \dfrac{1}{a_0}\end{Bmatrix}}_{\boldsymbol{B}}f(t)$$

→ 1st order state equation of dimension $n=l$

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Example 2: A l=2-nd order (dynamic) system with m=3 outputs and r=2 inputs

$$\begin{Bmatrix} \ddot{y}_1(t) \\ \ddot{y}_2(t) \\ \ddot{y}_3(t) \end{Bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{Bmatrix} \dot{y}_1(t) \\ \dot{y}_2(t) \\ \dot{y}_3(t) \end{Bmatrix} + \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \begin{Bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{Bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{Bmatrix} f_1(t) \\ f_2(t) \end{Bmatrix}$$

$\downarrow \Leftarrow$ augmentation of $[\dot{y}_1 \dot{y}_2 \dot{y}_3]^T$ *and* $[\ddot{y}_1 \ddot{y}_2 \ddot{y}_3]^T$ in state equation form

$$\underbrace{\begin{Bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \\ \ddot{y}_1 \\ \ddot{y}_2 \\ \ddot{y}_3 \end{Bmatrix}}_{\equiv \dot{\boldsymbol{x}}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ d_{11} & d_{12} & d_{13} & c_{11} & c_{12} & c_{13} \\ d_{21} & d_{22} & d_{23} & c_{21} & c_{22} & c_{23} \\ d_{31} & d_{23} & d_{33} & c_{31} & c_{32} & c_{33} \end{bmatrix}}_{\equiv \boldsymbol{A}:n\times n} \underbrace{\begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \end{Bmatrix}}_{\equiv \boldsymbol{x}} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ b_{11} & b_{12} \\ b_{21} & b_{12} \\ b_{31} & b_{12} \end{bmatrix}}_{\equiv \boldsymbol{B}:n\times r} \begin{Bmatrix} f_1(t) \\ f_2(t) \end{Bmatrix}$$

→ 1st order state equation of dimension $n=m\times l=3\times 2=6$

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Example 3: n-DOF vibration system (2nd order by Newton's 2nd Law)

$$\underbrace{\left(\boldsymbol{A}_0\frac{d^2}{dt^2}+\boldsymbol{A}_1\frac{d}{dt}+\boldsymbol{A}_2\right)}_{n\times n}\underset{n\times 1}{\boldsymbol{q}(t)}=\underset{n\times 1}{\boldsymbol{Q}(t)}$$

$$\downarrow\Leftarrow\begin{cases}\dot{\boldsymbol{q}}(t)=\dot{\boldsymbol{q}}(t)\\ \ddot{\boldsymbol{q}}(t)=\boldsymbol{A}_0^{-1}\left\{-\boldsymbol{A}_1\dot{\boldsymbol{q}}(t)-\boldsymbol{A}_2\boldsymbol{q}(t)+\boldsymbol{Q}(t)\right\}\end{cases}$$

$$\underbrace{\begin{Bmatrix}\dot{\boldsymbol{q}}(t)\\ \ddot{\boldsymbol{q}}(t)\end{Bmatrix}}_{\dot{\boldsymbol{X}}(t):2n\times 1}=\underbrace{\begin{bmatrix}0 & \boldsymbol{I}\\ -\boldsymbol{A}_0^{-1}\boldsymbol{A}_2 & -\boldsymbol{A}_0^{-1}\boldsymbol{A}_1\end{bmatrix}}_{\boldsymbol{A}:2n\times 2n}\underbrace{\begin{Bmatrix}\boldsymbol{q}(t)\\ \dot{\boldsymbol{q}}(t)\end{Bmatrix}}_{\boldsymbol{X}(t):2n\times 1}+\underbrace{\begin{Bmatrix}0\\ \boldsymbol{A}_0^{-1}\end{Bmatrix}}_{\boldsymbol{B}:2n\times n}\underset{\boldsymbol{f}:n\times 1}{\boldsymbol{Q}(t)}$$

$$\downarrow$$

$$\underset{2n\times 1}{\dot{\boldsymbol{x}}(t)}=\underset{2n\times 2n}{\boldsymbol{A}}\;\underset{2n\times 1}{\boldsymbol{x}(t)}+\underset{2n\times n}{\boldsymbol{B}}\;\underset{n\times 1}{\boldsymbol{f}(t)}$$

Note 1) 1st order state equation of dimension *2n*

Note 2) A_0, A_1, A_2 are normally symmetric but A is non-symmetric

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Ex. 1.7: Step response of a 2-nd order SISO system by means of the transient matrix & convolution integral

$$\ddot{q}+bq(t)=u(t), u(t):\ \text{unit step function}, q(0)=\dot{q}(0)=0$$

$$\downarrow \Leftarrow \begin{cases} \dot{q}=\dot{q} \\ \ddot{q}=-bq(t)+u(t) \end{cases}$$

$$\begin{Bmatrix} \dot{q} \\ \ddot{q} \end{Bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -b & 0 \end{bmatrix}}_{\boldsymbol{A}:2\times 2} \underbrace{\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}}_{\boldsymbol{X}(t):2\times 1} + \underbrace{\begin{Bmatrix} 0 \\ 1 \end{Bmatrix}}_{\boldsymbol{B}:2\times 1} u(t)$$

$$\downarrow$$

$$\dot{\boldsymbol{x}}=\boldsymbol{A}\boldsymbol{x}(t)+\boldsymbol{B}u(t):\ \boldsymbol{x}(t)=e^{\boldsymbol{A}t}\boldsymbol{x}(0)+\int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}u(\tau)d\tau$$

$$\downarrow \Leftarrow \begin{cases} e^{\boldsymbol{A}t}=\boldsymbol{I}+\sum_{k=1}^{\infty}\begin{bmatrix} 0 & 1 \\ -b & 0 \end{bmatrix}^k \frac{t^k}{k!} = \begin{bmatrix} 1-\frac{t^2b}{2!}+\frac{t^4b^2}{4!}-\ldots & t-\frac{t^3b}{3!}+\frac{t^5b^2}{5!}-\ldots \\ -tb+\frac{t^3b^2}{3!}-\frac{t^5b^3}{5!}-\ldots & 1-\frac{t^2b}{2!}+\frac{t^4b^2}{4!}-\ldots \end{bmatrix} = \begin{bmatrix} \cos\sqrt{b}t & \frac{1}{\sqrt{b}}\sin\sqrt{b}t \\ -\sqrt{b}\sin\sqrt{b}t & \cos\sqrt{b}t \end{bmatrix} \\ \boldsymbol{x}(0)=\boldsymbol{0} \end{cases}$$

$$\boldsymbol{x}(t)=\int_0^t \begin{bmatrix} \cos\sqrt{b}(t-\tau) & \frac{1}{\sqrt{b}}\sin\sqrt{b}(t-\tau) \\ -\sqrt{b}\sin\sqrt{b}(t-\tau) & \cos\sqrt{b}(t-\tau) \end{bmatrix} \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} u(\tau)d\tau$$

$$\downarrow$$

$$q(t)=\int_0^t \frac{1}{\sqrt{b}}\sin\sqrt{b}(t-\tau)d\tau=\frac{1}{b}\cos\sqrt{b}(t-\tau)\Big|_0^t=\frac{1}{b}\left(1-\cos\sqrt{b}t\right)=s(t):\ \text{unit step response}$$

: Great !

In-situ Quiz in Lecture 3: Derive the impulse response g(t) when u(t) is delta function using the Transition matrix and compare it to the Laplace Transformation solution. Send your answer to CLASSUM!

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Ex. 1.8: Transition matrix of the 2-nd order SISO system in Ex. 1.7 versus Laplace Transform solution

$$\ddot{q} + bq(t) = f(t)$$

$$\downarrow \Leftarrow \begin{cases} \dot{q} = \dot{q} \\ \ddot{q} = -bq(t) + f(t) \end{cases}$$

$$\begin{Bmatrix} \dot{q} \\ \ddot{q} \end{Bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -b & 0 \end{bmatrix}}_{\boldsymbol{A}:2\times 2} \underbrace{\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}}_{\boldsymbol{x}(t):2\times 1} + \underbrace{\begin{Bmatrix} 0 \\ 1 \end{Bmatrix}}_{\boldsymbol{B}:2\times 1} f(t)$$

$$\downarrow$$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}f(t) \left[\boldsymbol{x}(t) = \underbrace{e^{\boldsymbol{A}t}}_{\equiv \boldsymbol{\Phi}(t)} \boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)} \boldsymbol{B}f(\tau)\, d\tau \right]$$

$\downarrow \Leftarrow$ Laplace transform for initial condition $\boldsymbol{x}(0)$ and $f(t) = 0$

$$s\boldsymbol{x}(s) - \boldsymbol{x}(0) = \boldsymbol{A}\boldsymbol{x}(s) + \boldsymbol{B}f(s) : [s\boldsymbol{I} - \boldsymbol{A}]\boldsymbol{x}(s) = \boldsymbol{x}(0) : \boldsymbol{x}(s) = [s\boldsymbol{I} - \boldsymbol{A}]^{-1}\boldsymbol{x}(0) = \begin{bmatrix} s & -1 \\ b & s \end{bmatrix}^{-1} \boldsymbol{x}(0)$$

$$\boldsymbol{x}(s) = \underbrace{\begin{bmatrix} \dfrac{s}{s^2+b} & \dfrac{1}{s^2+b} \\ \dfrac{-b}{s^2+b} & \dfrac{s}{s^2+b} \end{bmatrix}}_{\equiv \boldsymbol{\Phi}(s) = L[\boldsymbol{\Phi}(t)] = L[e^{\boldsymbol{A}t}]} \boldsymbol{x}(0)$$

$\downarrow \Leftarrow$ Inverse Laplace transform

$$\boldsymbol{x}(t) = L^{-1}[\Phi(s)]\boldsymbol{x}(0) = \begin{bmatrix} L^{-1}\left[\dfrac{s}{s^2+b}\right] & L^{-1}\left[\dfrac{1}{s^2+b}\right] \\ L^{-1}\left[\dfrac{-b}{s^2+b}\right] & L^{-1}\left[\dfrac{s}{s^2+b}\right] \end{bmatrix} \boldsymbol{x}(0) = \underbrace{\begin{bmatrix} \cos\sqrt{b}t & \dfrac{1}{\sqrt{b}}\sin\sqrt{b}t \\ -\sqrt{b}\sin\sqrt{b}t & \cos\sqrt{b}t \end{bmatrix}}_{=\boldsymbol{\Phi}(t)} \boldsymbol{x}(0) = e^{\boldsymbol{A}t}\boldsymbol{x}(0)$$

: Great consistency!

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(Pb. 1.41) Impulse response of a damped 1-DOF system by the transition matrix (Hints)

$$\ddot{q}+2\dot{q}+25q(t)=\delta(t)\left[2\varsigma\omega_n=2,\omega_n^2=25,f(t)=\delta(t)\right]$$

$$\downarrow\Leftarrow\begin{cases}\dot{q}=\dot{q}\\ \ddot{q}=-2\varsigma\omega_n\dot{q}-\omega_n^2q(t)+f(t)\end{cases}$$

$$\begin{Bmatrix}\dot{q}\\ \ddot{q}\end{Bmatrix}=\underbrace{\begin{bmatrix}0 & 1\\ -\omega_n^2 & -2\varsigma\omega_n\end{bmatrix}}_{\boldsymbol{A}:2\times2}\underbrace{\begin{Bmatrix}q\\ \dot{q}\end{Bmatrix}}_{\boldsymbol{x}(t):2\times1}+\underbrace{\begin{Bmatrix}0\\ 1\end{Bmatrix}}_{\boldsymbol{B}:2\times1}f(t)$$

$$\downarrow$$

$$1)\boldsymbol{x}(t)=\underbrace{e^{\boldsymbol{A}t}}_{\equiv\boldsymbol{\Phi}(t)}\boldsymbol{x}(0)+\int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}f(\tau)d\tau\Leftarrow\begin{cases}e^{\boldsymbol{A}t}=\boldsymbol{I}+\sum_{k=1}^{\infty}\begin{bmatrix}0 & 1\\ -\omega_n^2 & -2\varsigma\omega_n\end{bmatrix}^k\frac{t^k}{k!}=[?]\text{: Analytically or numerically computed}\\ \boldsymbol{x}(0)=\begin{Bmatrix}0\\ 1\end{Bmatrix}\ \&\ f(t)=0\end{cases}$$

: Refer to Ex. 1.7

$$2)\boldsymbol{x}(s)=\left[s\boldsymbol{I}-\boldsymbol{A}\right]^{-1}\boldsymbol{x}(0)=\begin{bmatrix}s & -1\\ \omega_n^2 & s+2\varsigma\omega_n\end{bmatrix}^{-1}\boldsymbol{x}(0)=\underbrace{\begin{bmatrix}\frac{s+2\varsigma\omega_n}{s^2+2\varsigma\omega_ns+\omega_n^2} & \frac{1}{s^2+2\varsigma\omega_ns+\omega_n^2}\\ \frac{-\omega_n^2}{s^2+2\varsigma\omega_ns+\omega_n^2} & \frac{s}{s^2+2\varsigma\omega_ns+\omega_n^2}\end{bmatrix}}_{\equiv\boldsymbol{\Phi}(s)=L[\boldsymbol{\Phi}(t)]=L[e^{\boldsymbol{A}t}]}\boldsymbol{x}(0)$$

$\downarrow\Leftarrow$ Inverse Laplace transform: Refer to Ex. 1.8

$$3)\boldsymbol{x}(t)=e^{\boldsymbol{A}t}\boldsymbol{x}(0)\Leftarrow e^{\boldsymbol{A}t}=\text{expm}(\boldsymbol{A})$$ in matlab: Numerically

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Highlight 3

- Any order of linear MIMO systems can be formulated into 1-st order n-dimensional state equation.

- The **transition matrix** is an impulse response of the system determining the response to input.

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Supplementary (optional):

1.10 Discrete-Time Systems
(z-transform of dynamical systems)

Note: this additional material is for those who want to learn more about discrete-time systems which are frequently met in the digital computation procedures. The z-transform is the key process to convert continuous information to discrete version.

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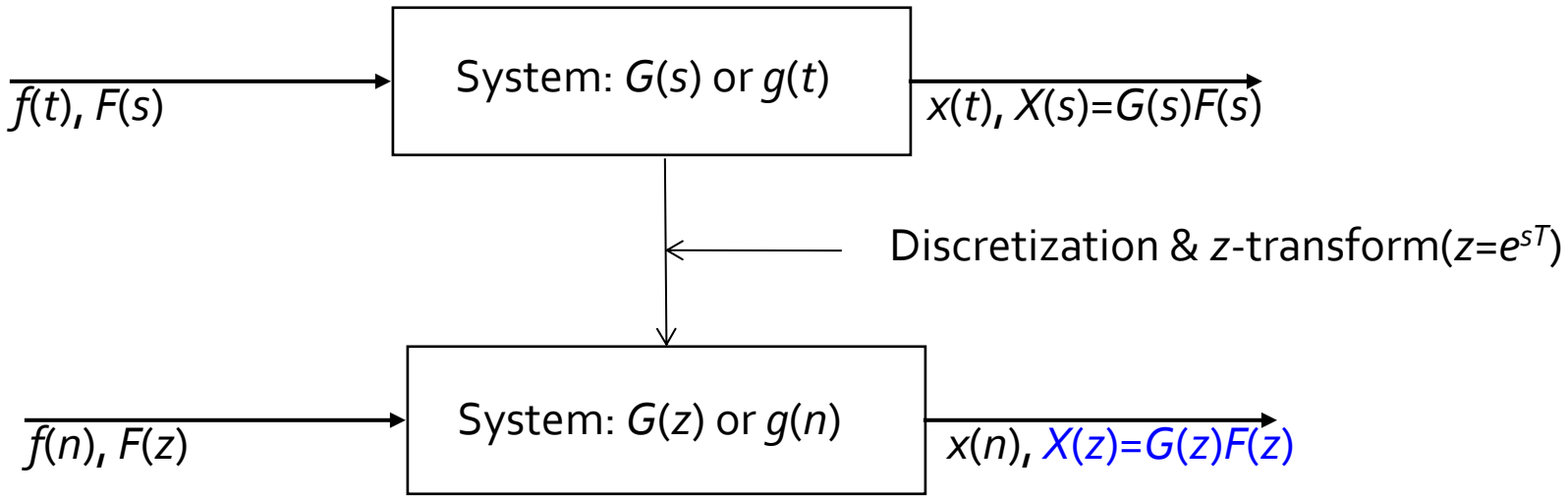
1.10 Discrete-Time Systems



$$X(s) \equiv \int_0^\infty x(t)e^{-st}dt : \text{ Laplace T.}$$

$$\downarrow \Leftarrow x(n) = x(t)\big|_{t=nT}, z \equiv e^{sT}$$

$$X(z) \equiv T\sum_{n=0}^{\infty} x(n)z^{-n} : \ z-\text{T.}$$

$$\downarrow$$

$$\Leftrightarrow X(z) = F(z)G(z)$$

Discrete-time impulse response: $g(k) \equiv g(t)\big|_{t=kT}$

Convolution summation: $x(n) = x(t)\big|_{t=nT} = \int_0^t f(\tau)g(t-\tau)d\tau\Big|_{t=nT}$

$$= T\sum_{k=0}^{n} f(k)g(n-k) = T\sum_{k=0}^{n} f(n-k)g(k)$$

Cf) Discretization versus quantization

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Discrete-Time Response in Recursive Form in Terms of Transition Matrix

$$\underset{n\times 1}{\dot{\boldsymbol{x}}(t)} = \underset{n\times n}{\boldsymbol{A}}\,\underset{n\times 1}{\boldsymbol{x}(t)} + \underset{n\times r}{\boldsymbol{B}}\,\underset{r\times 1}{\boldsymbol{f}(t)}$$

$\downarrow\Leftarrow$ Solution in terms of continuous transition matrix

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau$$

$\downarrow\Leftarrow$ Sampling at $t = kT$ & $(kT+T)$

$$\boldsymbol{x}(kT) = e^{\boldsymbol{A}kT}\boldsymbol{x}(0) + \int_0^{kT} e^{\boldsymbol{A}(kT-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau$$

$$\boldsymbol{x}(kT+T) = e^{\boldsymbol{A}(kT+T)}\boldsymbol{x}(0) + \int_0^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau = e^{\boldsymbol{A}(kT+T)}\boldsymbol{x}(0) + \int_0^{kT} e^{\boldsymbol{A}(kT+T-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau + \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau$$

$$= e^{\boldsymbol{A}T}\left[\underbrace{e^{\boldsymbol{A}kT}\boldsymbol{x}(0) + \int_0^{kT} e^{\boldsymbol{A}(kT-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau}_{\boldsymbol{x}(kT)}\right] + \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau$$

$$\downarrow\Leftarrow \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)}\boldsymbol{B}\boldsymbol{f}(\tau)d\tau \cong \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)}d\tau\boldsymbol{B}\boldsymbol{f}(kT)$$

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$$\downarrow$$

$$\boldsymbol{x}(kT+T) \cong e^{\boldsymbol{A}T}\boldsymbol{x}(kT) + \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)} d\tau \boldsymbol{B}\boldsymbol{f}(kT)$$

$$\downarrow \Leftarrow \int_{kT}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)} d\tau \underset{\sigma = kT+T-\tau}{\overset{\uparrow}{=}} \int_{T}^{0} e^{\boldsymbol{A}\sigma}(-d\sigma) = \int_{0}^{T} e^{\boldsymbol{A}\sigma} d\sigma = \boldsymbol{A}^{-1}\left(e^{\boldsymbol{A}T} - \boldsymbol{I}\right)$$

$$= \underbrace{e^{\boldsymbol{A}T}}_{\equiv \boldsymbol{\Phi}:\, n\times n \text{ constant}} \boldsymbol{x}(kT) + \underbrace{\boldsymbol{A}^{-1}\left(e^{\boldsymbol{A}T} - \boldsymbol{I}\right)\boldsymbol{B}}_{\equiv \boldsymbol{\Gamma}:\, n\times r \text{ constant}} \boldsymbol{f}(kT)$$

$$\underset{n\times 1}{\boldsymbol{x}(k+1)} = \underset{n\times n}{\boldsymbol{\Phi}}\, \boldsymbol{x}(k) + \underset{n\times r}{\boldsymbol{\Gamma}}\, \underset{r\times 1}{\boldsymbol{f}(k)}$$

Cf) $\boldsymbol{x}(k+1) = e^{\boldsymbol{A}(kT+T)}\boldsymbol{x}(0) + \int_{0}^{kT+T} e^{\boldsymbol{A}(kT+T-\tau)} \boldsymbol{B}\boldsymbol{f}(\tau) d\tau$ (See p. 9 of L2)

$$= e^{\boldsymbol{A}(kT+T)}\boldsymbol{x}(0) + T \underbrace{\sum_{n=0}^{k+1} e^{\boldsymbol{A}T(k+1-n)} \boldsymbol{B}\boldsymbol{f}(n)}_{\text{convolution summation over } (k+1) \text{ times}}$$

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